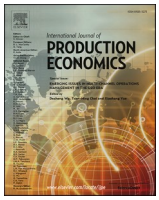
Int. J. Production Economics 230 (2020) 107890



Contents lists available at ScienceDirect

International Journal of Production Economics

journal homepage: http://www.elsevier.com/locate/ijpe





When should the e-tailer offer complimentary return-freight insurance ?

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ARTICLE INFO

Keywords:
Product returns
Return-freight insurance
E-commerce
Game theory

ABSTRACT

Consumers have recently become very concerned about the role of return-freight insurance (RI) in the e-commerce platform, where it appears prior to making the online purchase decision. In an effort to encourage consumers to buy products, some e-tailers will offer complimentary return-freight insurance (CRI), and it has become an established and recognized practice. Some e-tailers have stopped offering this resource, however, and offer consumers the opportunity to purchase RI. To examine an e-tailer's RI strategy, we consider a market consisting of a manufacturer and an e-tailer. Three alternative scenarios are investigated: 1) Scenario N where there is no RI; 2) Scenario C where consumers may purchase RI; and 3) Scenario S where the e-tailer offers CRI. We find that the introduction of RI increases the price level, and it does not necessarily benefit the manufacturer and the e-tailer. The RI premium incurred by the e-tailer and the return-freight compensation play a critical role in selecting the RI strategy. When the former is low and the latter is high, the e-tailer is willing to offer CRI. Meanwhile, the manufacturer has the incentive to bear the RI premium for the e-tailer.

1. Introduction

With the rapid development of mobile technology, online retailing has increased significantly in recent years. The China Internet Network Information Center notes that, in 2018, the number of online shoppers in China increased by 14.4% when compared with the preceding year, reaching 610 million in the process (CNNIC, 2019). In 2019, China's online sales increased by 16.5% compared with 2018, reaching 10.6 trillion RMB, accounting for 25.8% of total retail sales.[1] Although 75.2% of total retail sales still take place in stores, data shows that the online sales will account for more than 50% of the total retail sales in the next decade.[2]

In practice, product returns have come to play an increasingly important role in online retailing. Many consumers hesitate to shop online as they cannot have a "touch and feel" experience before purchasing (Ofek et al., 2011), with the consequence that products, and especially experience products, such as clothes and shoes (Information Week, 2013), often diverge from prior expectations (Mostard and Teunter, 2006). The National Retail Federation observes that, in 2017, the average product return rate in the United States was 10% (Retail, 2017); however, only about 5% of these returns were defective and 95% occurred for other reasons (Khouja et al., 2019). As the most popular shopping festival in China, "Double 11" not only creates high sales for e-tailers, but also leads to a high return rate. But the main reason for returns is not the product defect.[3] E-tailers have sought to reduce consumers' risk perception of online shopping through a variety of return policies that manage consumer returns, of which the Money-back guarantee (MBG) has emerged as the most popular (Davis et al., 1995; Akcay et al., 2013). This policy allows consumers to return products to e-tailers and receive full refunds. For example, e-tailers in *Taobao.com* and *JD.com* have long offered a seven-day "no question asked" full-refund return strategy. However, this lenient approach does not entirely resolve potential consumer concerns as e-tailers that make this concession generally require consumers to bear return-freight fees, unless products are defective. This is particularly problematic because, as has already been noted, failures to meet consumers' expectations, not product defects, are the main reason for online returns. The reverse logistics fee caused by return (return-freight fee) has therefore become a significant impediment to consumers when purchasing online (Li et al., 2018b).

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[1] http://m.news.cctv.com/2020/01/17/ARTI7bhk6aj8yhAH6d2LbRj6200117.shtml accessed on Apr 27, 2020.
[2] http://www.100ec.cn/detail–6519277.html accessed on Apr 27, 2020.
[3] http://www.100ec.cn/detail–6219758.html accessed on Apr 27, 2020.

https://doi.org/10.1016/j.ijpe.2020.107890
Received 21 December 2019; Received in revised form 24 July 2020; Accepted 8 August 2020
Available online 12 August 2020
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To reduce the adverse impact of return-freight fees and promote sales, return-freight insurance (RI) is born in China. If a consumer returns an insured product, the insurance company then compensates the consumer for the return-freight fee. Consumers confronted by the uncertainty of matching purchased products to their own expectations (product fit) have welcomed RI. Data shows that during the period October 2013–December 2016, the Chinese company Zhong An sold 7.2 billion RI, with 492 million consumers benefitting as a result.[4] In practice, both consumers and e-tailers are eligible to buy RI, but the latter has the preemptive right and RI can only be purchased once. Put differently, consumers can purchase RI on their own initiative only when e-tailers do not purchase it. In theory, when e-tailers buy RI, they will provide complimentary return-freight insurance (CRI) to consumers when selling products. In practice, we can identify this in ERKE's official flagship store in *JD.com*[5] and that of adidas in *Tmall.com*.[6] From the perspective of e-tailers, CRI improves consumer experience by reducing cost of having to return products to obtain more orders; however, CRI requires e-tailers to insure every product sold, and low return costs caused by the RI compensation may leads to more returns (Gu and Tayi, 2015; Geng et al., 2017a).

As the partner in the supply chain, the manufacturer has some "skin in the game" from the e-tailer's RI strategy. Offering CRI helps to increase consumers' willingness-to-pay, and this benefits the manufacturer. It will consequently most likely be in the manufacturer's interest to bolster the e-tailer's incentive to offer CRI. To this end, the manufacturer has a clear motivation to offer the e-tailer an incentive scheme, such as a cost-sharing contract (Ghosh and Shah, 2015; Niu et al., 2016).

In summary, the focus of this paper is to explore and understand the potential basis for the e-tailer to select the RI strategy. The existing literature does not provide a sufficient response and it will therefore be necessary to undertake further investigation. In particular, we propose to focus on the following key questions:

1) What is the impacts of RI on the manufacturer and the e-tailer?
2) What are the key factors that affect the RI strategy?
3) When should the e-tailer adopt the CRI strategy?

In seeking to answer these questions, we develop a two-stage supply chain including a manufacturer and an e-tailer where the former is the Stackelberg leader and the latter is the follower. Note that the e-tailer here is a third-party retailer who sells products through a marketplace (such as *Taobao.com* and *Tmall.com*). In practice, the e-tailer has the priority to purchase RI. If the e-tailer does not purchase it, consumers can get such an opportunity. We focus on three scenarios: 1) A scenario where there is no RI (Scenario N); 2) A scenario where consumers may purchase RI (Scenario C); 3) A scenario where the e-tailer offers CRI (Scenario S). Note that in Scenario C, consumers voluntarily choose whether to purchase RI, which means that some consumers who purchase products may not purchase it. In this paper, we first build the profit-maximization models of the manufacturer and the e-tailer in each scenario, and then obtain the optimal results which include the wholesale price, the retail price, and the profits. We then compare the optimal results of Scenario N with those of Scenario C and Scenario S, respectively, and this enables us to identify the impacts of RI on the manufacturer and the e-tailer. Finally, a comparison of Scenarios S and C enables us to obtain the optimal RI strategy.

The main contributions of our work are as follows. First, we investigate the e-tailer's RI strategy in a context where RI is open to both the e-tailer and consumers. To the best of our knowledge, the existing literature only focuses on the situation that the e-tailer has the opportunity to purchase RI. Second, we obtain the conditions of the e-tailer's optimal RI strategy and find that they are closely related to the RI premium incurred by the e-tailer and the return-freight compensation. Our results show that the low RI premium and the high return-freight compensation make the e-tailer tend to offer CRI. This can provide useful decision support to the e-tailer. Third, we explore the possibility that the manufacturer could offer a cost-sharing contract that would encourage the e-tailer to offer CRI, which can provide practical management insights for supply chain members in practice.

This paper is organized as follows. Section 2 summarizes the relevant literature. Notations and the problem description are given in Section 3. The impact of RI on the manufacturer and the e-tailer is investigated in Section 4. Section 5 examines the criteria for the e-tailer to choose a RI strategy and the impacts of several important parameters on this criteria. Conclusions, managerial insights and future research directions are provided in Section 6.

2. Literature review

Our work primarily relates to two streams of research: MBGs and consumer returns, and RI. Below, we review each of these streams of research and demonstrate how they interact with our research.

Consumer returns are a common phenomenon in the retailing industry. Various return policies that protect consumers from unsatisfactory purchases are discussed in the literature (e.g., Shieh, 1996; Su, 2009). Some studies focus on the effect of return strategies on firms' operations strategies, including pricing, order quantity and supply chain coordination (Chen and Bell, 2009; Xiao et al., 2010; Ruiz-Benitez and Muriel, 2014; Xu et al., 2015). The MBG, as the most popular return strategy that commits retailers to offer full refunds to consumers for any reason (Davis et al., 1995), has been widely studied in the operations management literature. Moorthy and Srinivasan (1995) study how MBGs convey a quality signal to consumers in the setting where it is not possible for them to ascertain high or low quality before purchasing. McWilliams (2012) develops a duopoly model to investigate the impact of MBGs on two competing retailers, who sell high- and low-quality products, respectively. The research shows that MBGs hurt the former and benefit the latter. Akcay et al. (2013) study the retailer's decisions relating to new product price, procurement quantity, refund amount, and the price of returned products when they are sold as open-box items. They find that selling with the MBG increases sales and profit. Chen and Grewal (2013) study a two-stage supply chain consisting of a manufacturer and two retailers who compete in Nash or Stankelberg games. One retailer has a good reputation and provides the MBG; while the other is a new entrant who needs to decide whether to offer this concession. They find that if the transportation cost is low, it is optimal for the new entry retailer not to provide the MBG. Hsiao and Chen (2014) compare the profitability of the MBG and the hassle-free strategy. They find the former to be more profitable even if high-valuation consumers encounter lower hassle costs than low-valuation consumers. Yang et al. (2017) investigate a supply chain consisting of a retailer and two manufacturers (a well-known brand and a new brand), which provides the retailer with the option of selling to one or both. The results show the MBG enhances retailer's profit and expands the overall market. Chen and Chen (2017) develop a model to study the return strategy of a retailer with two channels (who is confronted by the dilemma of whether to offer the MBG in each channel). They conclude that the retailer should offer an MBG in a channel as long as the net salvage value of the returned product is positive in that channel. Li et al. (2018a) discuss the decision of a retailer who sells two brands of products on return policies, and consider the question of whether to provide the MBG. They show that the MBG mitigates price competition between the two brands and increases the retailer's profit. In practice, MBGs cannot completely address consumers' concerns, and the return-freight fee therefore continues to impede consumers who purchase online. According to the observation, e-tailers offering MBGs can decide again

[4] http://www.100ec.cn/home/detail–6421595.html accessed on Apr 27, 2020.
[5] https://erkeyd.jd.com/accessed on Dec 21, 2019.
[6] https://adidas.tmall.com/search.htm accessed on Dec 21, 2019.

whether to offer CRI for consumers who can get a certain amount of compensation for the return-freight fee after returning. Our study is based on the MBG setting. Specifically, the MBG is the premise of introducing RI. Being different from the above research, our objective is to examine the value of RI and whether the e-tailer should offer CRI as a better return service.

There is limited literature on RI. Geng et al. (2017a) utilize the signaling theory to investigate whether the e-tailer can convey the product quality signal to consumers through offering CRI. They consider a market consisting of an e-tailer who can choose to sell low-quality or high-quality products and a group of consumers who do not know the product quality in advance. They find that 1) the e-tailer who sells low-quality products will not offer CRI, and 2) for the e-tailer who sells high-quality products, CRI can be used to convey the impression of a high-quality product to consumers. Different from the traditional insurance premium analysis framework, Geng et al. (2017b) study the insurance company's optimal RI pricing strategy, including the insurance premium and the compensation, by merging the characteristic of e-commerce (such as the uncertainty of product fit in online shopping). In addition, they also examine the impacts of the uncertainty of product fit on both fees. The results show that: 1) when the uncertainty of product fit is high, the compensation should be fixed as the return-freight fee, while the insurance premium changes with the uncertainty of product fit, and 2) when the consumers' sensitivity to the uncertainty of product fit is moderate, the optimal insurance premium goes up with the increase of the uncertainty of product fit, but the optimal compensation goes down to the opposite direction. Li et al. (2018b) focus on exploring the impacts of RI on the online platform (such as *Taobao.com*) and two third-party retailers (both of them offer CRI to consumers). Some important results are revealed. First, the provision of RI usually induces retailers to increase their prices. Second, whether RI benefits retailers and the online platform is not monotonous, which depends on the basic return rate and the degree of product differentiation. Specifically, when the retailers have the same basic return rate, RI can increase the online platform's profit, but has no impact on the retailers; when their basic return rate are different, RI is only beneficial to the retailer with the high basic return rate and only beneficial to the online platform when the product differentiation is high. However, these studies focus on the scenario where the e-tailer offers CRI, and ignore the scenario where consumers buy RI. Driven by the observable examples, both dimensions are considered in our work and this enriches the literature.

3. Notations and problem description

To make it easier to track the problem description, relevant notations are shown in Table 1, where $i = N, C, S$ stand for Scenario N, Scenario C, and Scenario S, respectively.

This paper investigates a two-stage supply chain consisting of a manufacturer (he) who is leader in a Stackelberg game and an e-tailer (she) who is the follower. The e-tailer buys products from the manufacturer at w_i, and then sells them to consumers at p_i. Consumers cannot experience products when shopping online, and the products they receive may not match prior expectations. We assume that when a product does not match a consumer's expectations, the consumer will return the product to the e-tailer and receive a full-refund p_i. For each product returned, we assume that the salvage value proportion is $\theta \in (0, 1]$. For the e-tailer, the value of a unsold product is w_i, then the salvage value of each product returned is θw_i.

In practice, if the product itself does not have defects, consumers generally need to pay the return-freight fee s when returning (Chen and Chen, 2017; Li et al., 2018b). To improve consumer experience and reduce the adverse impact of return-freight fees, the e-tailer can choose to spend f_s to purchase RI for consumers, that is, to offer CRI for consumers. Note that the RI premium incurred by the e-tailer is related to her historical average return rate of products sold. Generally, the higher the return rate is, the higher the RI premium that she needs to pay is. If consumers purchase the insured product, they can obtain the return-freight compensation r ($r \leq s$) provided by the insurance company when returning. In addition, if the e-tailer does not provide this resource, consumers can also consider purchasing RI. Based on this, we focus on the following three scenarios, as shown in Fig. 1.

1) Scenario N where there is no RI. In this (benchmark) scenario, if consumers are not satisfied with products they receive, they will need to pay the return-freight fee s to return the product to the e-tailer.
2) Scenario C where consumers may purchase RI. Before placing orders online, consumers have to decide whether to spend f_c to buy RI if the e-tailer does not offer CRI. When returns occur, consumers who do not purchase RI need to fully bear the return-freight fee s, while consumers who purchase RI only need to pay $s - r$. Note that the latter have to pay f_c, irrespective of the occurrence of returns.
3) Scenario S where the e-tailer offers CRI. In this scenario, all consumers can obtain the return-freight compensation r. The manufacturer in the supply chain appears as a partner of the e-tailer, and the RI strategy endows him with some "skin in the game". Consequently, it is in his interest to provide an added incentive for the e-tailer to offer CRI. One way to achieve this would be to offer a cost-sharing contract to the e-tailer. In the case of the RI premium f_s, we assume the proportion the e-tailer would share with the manufacturer, as indicated in the contract, is $\alpha \in [0, 1]$.

The decision sequence establishes that the manufacturer first announces α (Scenario S only. In practice, because the manufacturer has no responsibility to share the RI premium, the proportion should be determined by him); the manufacturer then also determines w_i; finally, the e-tailer determines p_i.

To match the practice, our model captures the consumer heterogeneity with respect to two key features, namely, the consumer valuation of a product and the RI premium f_c. First, our model allows for the consumer heterogeneity for the product value, denoted by v, where v is a random variable with support $[0, 1]$. Second, our model allows for the consumer heterogeneity with respect to the RI premium f_c, which is also a random variable with support $[0, 1]$. In practice, the RI premium incurred by a consumer depends on the historical risk probability of RI (the number of successful claims/the number of successful insurances), which is different from the premium incurred by the e-tailer. Therefore, the RI premium varies for different consumers.

Next, we provide two comparisons between Scenarios N and C, and Scenarios N and S. These comparisons can help us identify the impacts of introducing RI on the decisions of supply chain members. Then, in Section 5, we compare Scenarios C and S to analyze the optimal RI strategy.

Table 1
Notations.

Notations	Descriptions
w_i	Wholesale price in Scenario i (decision variable)
p_i	Retail price in Scenario i (decision variable)
α	Proportion of the RI premium borne by the e-tailer (decision variable)
v	The consumer's valuation of the product
m	The probability of product fit ($0 < m < 1$)
c	Per-unit production cost
s	Per-unit return-freight fee (incurred by a consumer)
r	Per-unit return-freight compensation (obtained by a consumer)
f_s	Per-unit return-freight insurance premium (incurred by an e-tailer)
f_c	Per-unit return-freight insurance premium (incurred by a consumer)
θ	Salvage value proportion

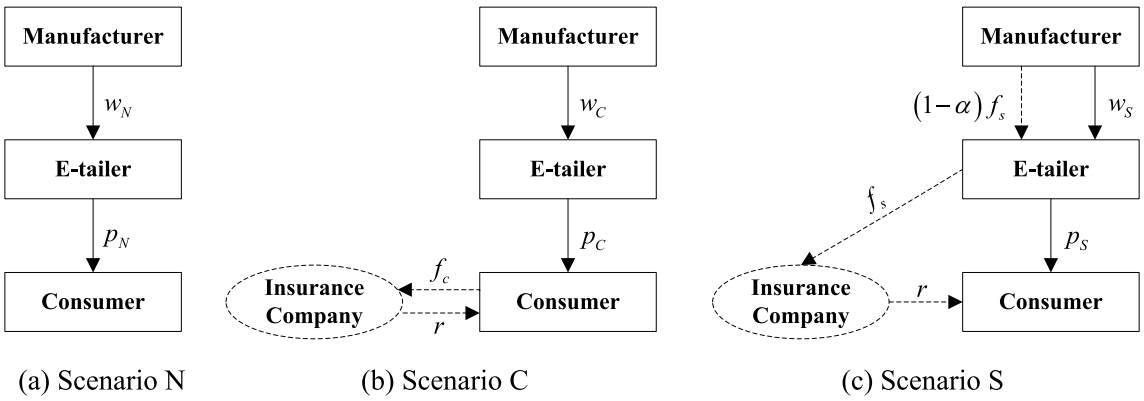


Fig. 1. Three scenarios: Scenarios N, C and S.

4. Impacts of return-freight insurance

In this section, we first consider the benchmark case where there is no RI (Scenario N). Then, we analyze two scenarios after introducing RI, i.e., Scenarios C and S. Comparing the latter two scenarios with Scenario N, we obtain the impacts of RI.

4.1. No return-freight insurance (Scenario N)

Referring to the existing research (Chen and Chen, 2017; Yang et al., 2017), a consumer's expected utility is $U_N = mv + (1-m)(p_N - s) - p_N$. The consumer will buy the product only if the expected utility is non-negative, i.e., $U_N \geq 0$, and this gives $v \geq p_N + \frac{1-m}{m}s$. The consumer utility in Scenario N is presented in Fig. 2. By calculating the region of $U_N \geq 0$, we obtain the demand $D_N = 1 - p_N - \frac{1-m}{m}s$.

The e-tailer's profit is

$$\Pi_N^R = D_N[mp_N + (1-m)\theta w_N - w_N], \tag{1}$$

and the manufacturer's profit is

$$\Pi_N^M = D_N(w_N - c). \tag{2}$$

According to the principle of profit maximization, it is easy to prove the following results by using backward induction.

Lemma 1. *In Scenario N, the manufacturer's optimal wholesale price is* $w_N^* = \frac{m-(1-m)s+\lambda c}{2\lambda}$, *the e-tailer's optimal retail price is* $p_N^* = \frac{3[m-(1-m)s]+\lambda c}{4m}$, *the demand is* $D_N^* = \frac{m-(1-m)s-\lambda c}{4m}$, *and profits are* $\Pi_N^{M*} = \frac{[m-(1-m)s-\lambda c]^2}{8m\lambda}$ *and* $\Pi_N^{R*} = \frac{[m-(1-m)s-\lambda c]^2}{16m}$, *where* $\lambda = 1 - \theta(1-m)$.

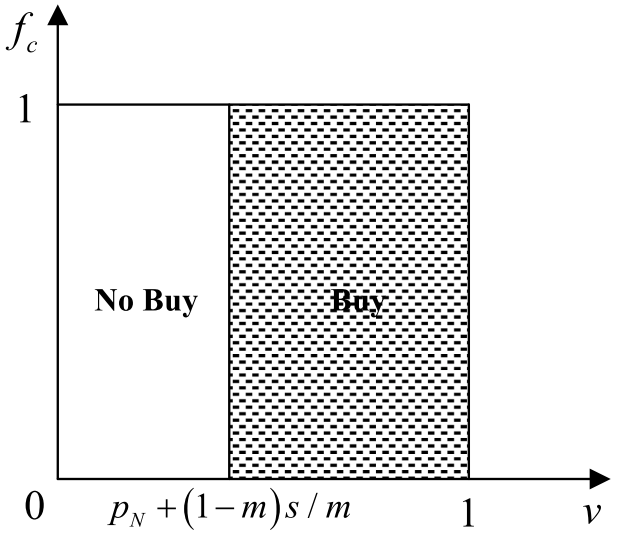


Fig. 2. Consumer purchasing preference in Scenario N.

Lemma 1 clarifies that both the wholesale and retail prices decrease as return-freight fees increase. The purchased products may not match consumers' expectations, and they therefore need to consider the adverse effect of returns when making purchase decisions. In this way, as return-freight fees increase, the manufacturer will reduce the wholesale price to entice the e-tailer to reduce the retail price, so as to achieve the purpose of improving consumers' willingness-to-pay. In addition, it is clear that both the wholesale price and the demand increase with the increase of salvage value proportion, while the retail price decreases with the increase of it. This is because a high salvage value proportion means that the e-tailer bears a low return loss and can "boldly" purchase, so that the manufacturer hopes to create more profits by raising the wholesale price. The low return loss encourages the e-tailer to obtain more demand by reducing the retail price. Intuitively, a low retail price will improve consumers' willingness-to-pay, and this can increase the demand.

4.2. Consumer purchasing RI (Scenario C)

We now examine the scenario where consumers choose to purchase RI independently, which also means that the e-tailer does not offer CRI. Similar to Scenario N, the expected utility of a consumer who only purchases the product is $U_{C1} = mv + (1-m)(p_C - s) - p_C$. If a consumer purchases both the product and RI, the expected utility becomes $U_{C2} = mv + (1-m)(p_C + r - s) - p_C - f_c$. The consumer will buy product if $U_{C1} \geq 0$ (i.e., $v \geq p_C + \frac{1-m}{m}s$). Similarly, the consumer will buy the product and RI if $U_{C2} \geq 0$ (i.e., $f_c \leq m(v - p_C) - (1-m)(s-r)$). On this basis, if $U_{C2} \geq U_{C1}$ (i.e., $f_c \leq (1-m)r$), consumers finally choose the RI option, otherwise, they only buy the product. The consumer utility in Scenario C is presented in Fig. 3. In regions A and B, consumers insure the products they buy; in region C, consumers only buy products. Calculating the regions A, B and C leads to the total demand, which is $D_C = 1 - p_C - \frac{1-m}{m}s + \frac{(1-m)^2r^2}{2m}$.

In addition, we can see from the region A that more consumers will try to buy products, which also implies that the introduction of RI helps the e-tailer expand the market. The additional demand is $D' = \frac{(1-m)^2r^2}{2m}$. It can be found that D' increases with the increase of return-freight compensation r, and decreases with the increase of probability of product fit m. This is because that a high return-freight compensation r can ease the adverse impact brought by the return-freight fee, then consumers' willingness-to-pay is improved; as the probability of product fit m increases, consumers' shopping risk is reduced, and then the attraction of the return-freight compensation r to consumers is reduced, which leads to the reduction of this part of demand.

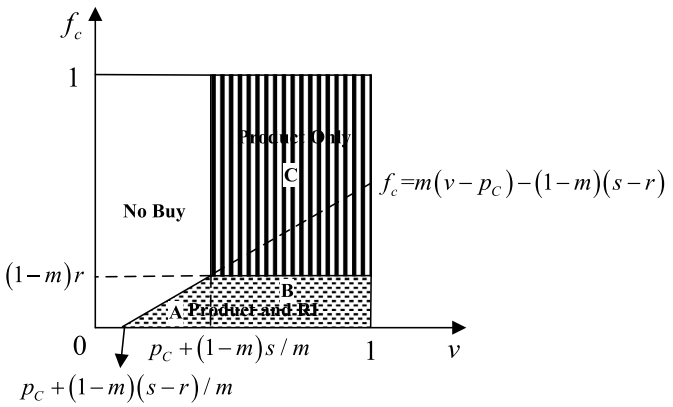


Fig. 3. Consumer purchasing preference in Scenario C.

In Scenario C, the profit expressions of supply chain members are consistent with Eqs. (1) and (2). By solving, we have the results summarized in Lemma 2.

Lemma 2. *In Scenario C, the manufacturer's optimal wholesale price is* $w_C^* = \frac{m-(1-m)s+\lambda c+(1-m)^2 r^2/2}{2\lambda}$, *the e-tailer's optimal retail price is* $p_C^* = \frac{3[m-(1-m)s]+\lambda c+3(1-m)^2 r^2/2}{4m}$, *the demand is* $D_C^* = \frac{m-(1-m)s-\lambda c+(1-m)^2 r^2/2}{4m}$, *and profits are* $\Pi_C^{M*} = \frac{[m-(1-m)s-\lambda c+(1-m)^2 r^2/2]^2}{8m\lambda}$ *and* $\Pi_C^{R*} = \frac{[m-(1-m)s-\lambda c+(1-m)^2 r^2/2]^2}{16m}$.

Consistent with the results in Lemma 1, Lemma 2 shows that the wholesale price, the retail price and the demand in Scenario C also decrease with the increase of return-freight fees. More importantly, we find that as the return-freight compensation increases, the wholesale price, the retail price and the demand will increase, which is intuitive. The return-freight compensation can reduce the shopping risk of consumers, which leads to a high willingness-to-pay. Meanwhile, it also improves the pricing space of the manufacturer and the e-tailer. As a result, supply chain members hope to earn more profits from the increased demand by increasing prices.

4.3. Offering CRI (Scenario S)

We discuss the scenario in which the e-tailer provides consumers with CRI in this section. In addition, we explore whether the manufacturer has the incentive to share the RI premium with the e-tailer through a cost sharing contract. We assume the proportion that the e-tailer and the manufacturer have to bear is $\alpha$and $1-\alpha$, respectively. Because the manufacturer has no responsibility to share the RI premium, the proportion should be determined by him. In particular, when $\alpha = 0$, the total premium is only borne by the manufacturer; when $\alpha = 1$, it is only borne by the e-tailer.

Unlike Scenarios N and C, consumers in Scenario S do not need to purchase RI to get the return-freight compensation. Then, the expected utility is $U_S = mv + (1-m)(p_S + r - s) - p_S$. The consumer will purchase the product if $U_S \geq 0$, which gives $v_S \geq p_S + \frac{1-m}{m}(s-r)$. The consumer utility in Scenario S is presented in Fig. 4. By calculating the region of $U_S \geq 0$, we obtain the demand $D_S = 1 - p_S - \frac{1-m}{m}(s-r)$.

The profit of the e-tailer is

$$\Pi_S^R = D_S[mp_S + (1-m)\theta w_S - w_S - \alpha f_s], \tag{3}$$

and the profit of the manufacturer is

$$\Pi_S^M = D_S[w_S - c - (1-\alpha)f_s] \tag{4}$$

Lemma 3 summarizes the optimal results in Scenario S.

Lemma 3. *In Scenario S, the manufacturer's optimal wholesale price is* $w_S^* = \frac{m-(1-m)(s-r)+\lambda c+\lambda f_s}{2\lambda}$, *the proportion that the e-tailer has to bear and her optimal retail price are respectively* $\alpha^* = 0$ *and* $p_S^* = \frac{3[m-(1-m)(s-r)]+\lambda c+\lambda f_s}{4m}$, *the demand is* $D_S^* = \frac{m-(1-m)(s-r)-\lambda c-\lambda f_s}{2\lambda}$, *and profits are* $\Pi_S^{M*} = \frac{\{m-(1-m)(s-r)-\lambda c-\lambda f_s\}^2}{8m\lambda}$ *and* $\Pi_S^{R*} = \frac{\{m-(1-m)(s-r)-\lambda c-\lambda f_s\}^2}{16m}$.

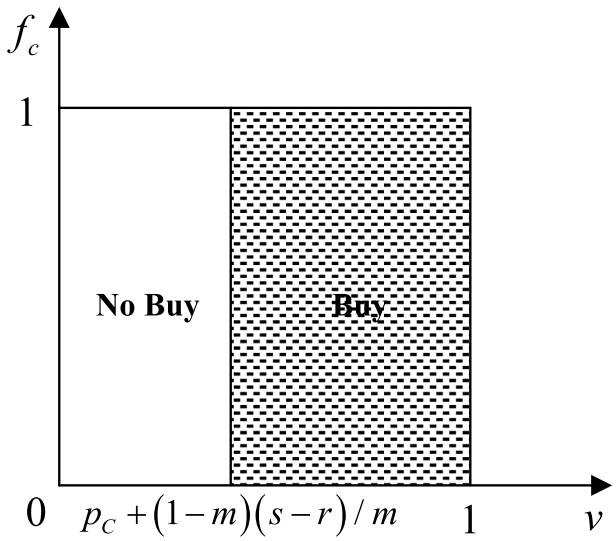


Fig. 4. Consumer purchasing preference in Scenario S.

Lemma 3 makes it straightforward to identify that the wholesale and retail prices are positively proportional to the RI premium incurred by the e-tailer. The perspective of cost allocation makes it clear that, as the manufacturer bears the total RI premium, he will transfer the cost to the e-tailer by raising the wholesale price; the e-tailer will then transfer the cost to consumers in the same way. The increase of retail price caused by the increase of RI premium leads to the decrease of demand, which can explain why the demand decreases with the increase of RI premium.

4.4. Comparison analysis

In the previous section, we get the results in three scenarios. By comparing Scenarios N and C, and Scenarios N and S, we now try to answer the question that the impacts of RI on the demand and supply chain members' profits. We mainly investigate how two key factors, namely, the RI premium incurred by the e-tailer and the return-freight compensation, affect the impacts of RI.

We first discuss the impact of RI on the demand. By Lemmas 1–3, we can obtain the following proposition. All proofs are provided in Appendix.

Proposition 1. $\frac{\partial D_N^*}{\partial m} > \frac{\partial D_C^*}{\partial m} > 0$. *When* $(1-\theta)f < r \leq s$, $\frac{\partial D_N^*}{\partial m} > \frac{\partial D_S^*}{\partial m} > 0$; *When* $0 < r \leq (1-\theta)f_s$, $\frac{\partial D_S^*}{\partial m} \geq \frac{\partial D_N^*}{\partial m} > 0$.

First of all, from Proposition 1, it is clear that in any scenario, the demand increases with the increase of probability of product fit, which is reasonable. This is because that as the probability of product fit increases, consumers can more ensure that the products will meet their expectations, and the adverse impact of the return-freight fee on consumers will be reduced. Thus, the demand will increase. Next, we also find some interesting results. The impact of the probability of product fit on the demand in Scenario C is always less than that in Scenario N, while the situation in Scenario S does not always show such a trend. The occurrence of above results is due to two opposite effects of RI on the demand. On the one hand, the provision of the return-freight compensation will lead to a decline in consumers' attention to the probability of product fit, because consumers' shopping risk is reduced. Then, consumers will be more "bold" to purchase products. On the other hand, the

e-tailer will increase the retail price ($p_S^* > p_N^*$), which makes consumers pay more attention to the probability of product fit, because they want to keep the products. Meanwhile, as the RI premium incurred by the e-tailer increases, the retail price increases in Scenario S ($\partial p_S^*/\partial f_s > 0$). Specifically, the effect of the return-freight compensation on reducing consumers' attention to the probability of product fit is dominant in Scenario C, then the impact of the probability of product fit on the demand in Scenario C is always less than that in Scenario N. In Scenario S, the positive effect caused by the RI premium incurred by the e-tailer is dominated by the negative effect caused by the return-freight compensation when the RI premium is low, and then the impact of the probability of product fit on the demand in Scenario S is less than that in Scenario N; when the RI premium is high, the effect of the RI premium on increasing consumers' attention to the probability of product fit is dominant in Scenario S, and then the impact of the probability of product fit on the demand in Scenario S is higher than that in Scenario N.

We compare the optimal prices in Scenarios N and C, and those in Scenarios N and S to identify the impacts of RI on the pricing strategy of supply chain members, as shown in Proposition 2.

Proposition 2. *The introduction of RI will raise the price level in the market. Mathematically,* $w_C^* > w_N^*$, $w_S^* > w_N^*$; $p_C^* > p_N^*$, $p_S^* > p_N^*$.

From Proposition 2, we find that the introduction of RI will urge the manufacturer to ask the e-tailer for a higher wholesale price. Meanwhile, the e-tailer will also charge consumers a higher retail price, which also verifies the research result of Li et al. (2018b) on the e-tailer's pricing strategy. The reasons are as follows. In Scenario C, consumers can voluntarily choose to purchase RI to mitigate their shopping risk. In this way, the e-tailer may hope to earn more profits from this voluntary behavior by increasing the retail price. Observing this, the manufacturer will also raise the wholesale price to create more profits for himself. In view of the price increase in Scenario S, on the one hand, the provision of the return-freight compensation can alleviate the adverse impact of a high retail price on consumers' willingness-to-pay; on the other hand, the increase of operational costs inevitably leads to a high price, including the wholesale and retail prices.

We then study the impact of RI on the profits of the manufacturer and e-tailer. Proposition 3 provides a clear answer.

Proposition 3. 1) *When the e-tailer does not offer CRI, we have:* $\Pi_C^{M^*} > \Pi_N^{M^*}$, $\Pi_C^{R^*} > \Pi_N^{R^*}$.

2) *When the e-tailer offers CRI, we have: if* $0 < f_s \le \tilde{f} = (1-m)r/\lambda$, $\Pi_S^{M^*} \ge \Pi_N^{M^*}$, $\Pi_S^{R^*} \ge \Pi_N^{R^*}$; *if* $(1-m)r/\lambda < f_s \le 1$, $\Pi_S^{M^*} < \Pi_N^{M^*}$, $\Pi_S^{R^*} < \Pi_N^{R^*}$.

Part 1) implies that the introduction of RI creates more profits for the manufacturer and the e-tailer even without offering CRI. Because of the RI option, some consumers can purchase RI to reduce their shopping risk, and then try to buy products (as shown in the region A in Fig. 3). Thus, RI helps to expand the consumer market ($D_C^* > D_N^*$), and supply chain members can also take advantage of this to earn more profits.

Part 2) reveals that the change of profits in Scenario S is not monotonous. When the RI premium incurred by the e-tailer is low, supply chain members can get more profits from the CRI strategy, otherwise, this strategy will damage their profits. As found in Section 4.3, the manufacturer is fully responsible for the RI premium incurred by the e-tailer. Then, a high RI premium will make the manufacturer increase the wholesale price, which will also prompt the e-tailer to set a high retail price, thus reducing consumers' willingness-to-pay. As a result, the relatively high RI premium will obviously reduce the enthusiasm of supply chain members for offering CRI.

Note that $\tilde{f}$ is inversely proportional to the probability of product fit, which means that a high probability of product fit will reduce the probability of offering CRI. The purpose of RI is to attract more potential consumers to buy products. However, as the probability of product fit increases, consumers can ensure that products will meet their expectations, with the consequence that the incentive effect of RI will not be significant. Therefore, with a high probability of product fit, the manufacturer and the e-tailer are not inclined to offer CRI.

5. Discussion

In this part, we mainly explore when the e-tailer should offer CRI. It should be emphasized that, different from the studies of Geng et al. (2017a) and Li et al. (2018b), i.e., the RI strategy is obtained by comparing the e-tailer's profits in the scenarios of offering CRI and no RI, we also examine the scenario where consumers purchase RI. To clearly analyze the strategic implication of CRI, we still focus on the two indicators: the RI premium incurred by the e-tailer and the return-freight compensation. In addition, we also analyze the impacts of other important parameters on the RI strategy, including the probability of product fit and the salvage value proportion.

5.1. Return-freight insurance strategy

To offer more sights, we first present the comparison results of retail prices and wholesale prices, which can reflect the impacts of RI strategy on the pricing strategy of supply chain members, as shown in Proposition 4.

Proposition 4. *For two strategies of RI, that is, offering or not offering CRI (RI is avaliable for consumers to purchase), the manufacturer and the e-tailer will set higher prices in the former strategy. Mathematically,* $w_S^* > w_C^*$, $p_S^* > p_C^*$.

Proposition 4 provides the comparison results of the optimal prices in Scenarios C and S. The results are monotonous, that is, the optimal wholesale and retail prices in Scenario S are higher than those in Scenario C, respectively. It also suggests that CRI improves the price level in the market with RI. This result is reasonable for the following reasons. Assuming that the retail prices in the two scenarios are the same, and we can clearly find that the demand in Scenario S is higher than that in Scenario C, as shown in Figs. 3 and 4. This also indirectly indicates that the e-tailer needs to handle more returns in Scenario S. The high loss of product value brought by the increase of returns will urge the e-tailer to ensure her profits by increasing the retail price. In addition, it can be found from Lemma 3 that in order to maximize his own interests, the manufacturer will bear all the RI premium incurred by the e-tailer. Therefore, due to the increased operational costs, the manufacturer has the incentive to increase the wholesale price to ensure his marginal profits, which will also induce the e-tailer to set a high retail price.

Below, we compare the profits of supply chain members in Scenarios C and S, and discuss whether the supply chain members can benefit from CRI. Proposition 5 answers this question.

Proposition 5. *When the RI premium incurred by the e-tailer is low and the return-freight compensation is high, the e-tailer is willing to offer CRI, and the manufacturer shows the same preference, otherwise they tend not provide this resource. Mathematically, when* $0 < f_s < (1-m)s[2-(1-m)s]/2\lambda$, *if* $0 < r \le \hat{r}$, $\Pi_C^{M^*} \ge \Pi_S^{M^*}$, $\Pi_C^{R^*} \ge \Pi_S^{R^*}$, *and if* $\hat{r} < r \le s$, $\Pi_S^{M^*} \ge \Pi_C^{M^*}$, $\Pi_S^{R^*} \ge \Pi_C^{R^*}$; *when* $(1-m)s[2-(1-m)s]/2\lambda < f_s \le 1$, $\Pi_C^{M^*} \ge \Pi_S^{M^*}$, $\Pi_C^{R^*} \ge \Pi_S^{R^*}$.

Proposition 5 reveals that the RI premium incurred by the e-tailer and the return-freight compensation will jointly affect the RI strategy, which is different from that Geng et al. (2017a) utilize the product quality to measure the e-tailer's motivation of offering CRI. When the RI premium is low and the return-freight compensation is high, CRI will benefit the manufacturer and the e-tailer, otherwise, it will be harm for them. On the one hand, CRI can reduce the shopping risk of all consumers, and then improve their willingness-to-pay. Note that this positive effect depends on the return-freight compensation. The higher the

return-freight compensation, the greater the positive effect. On the other hand, the e-tailer will charge a higher retail price (as shown in Proposition 3) after offering CRI, which then reduce consumers' willingness-to-pay. Meanwhile, as the RI premium increases, the retail price increases ($\partial p_S^*/\partial f_s > 0$). Therefore, the negative effect depends on the RI premium. In summary, when the RI premium is very high, the negative effect plays a dominating role; on the contrary, if it is not very high, the positive effect is dominant when the return-freight compensation is high, otherwise, the negative effect is dominant. Our findings provide executable management insights. For example, when the return-freight compensation is high, the e-tailer with the low historical average return rate (the RI premium paid by the e-tailer depends on her historical average return rate of products sold. The higher the historical average return rate, the higher the RI premium) should offer CRI.

5.2. Sensitivity analysis

In this section, we explore the impacts of the probability of product fit and the salvage value proportion on the RI strategy. As illustrated in Figs. 5 and 6, parameters of different sizes correspond to different threshold lines. Below each threshold line, the optimal RI strategy is to offer CRI, and above each threshold line, it is better to let consumers purchase RI. We then analyze the impacts of these parameters by illustrating how this threshold line moves with them. Without loss of generality, we set the basic parameter values as $m = 0.8, s = 0.08, c = 0.3, \theta = 0.85$. Note that the parameters in the following numerical experiments will follow this basic setting unless otherwise stated.

Fig. 5 shows the impact of the probability of product fit on the RI strategy. We can find that as the probability of product fit decreases, the threshold line moves up. This implies that the e-tailer is more likely to offer CRI. The reduction of probability of product fit means that consumers face the high shopping risk, and the provision of the return-freight compensation can ease the concerns of all consumers. As a result, offering CRI will become the better choice for the e-tailer.

From Fig. 6, we observe that as the salvage value proportion increases, the optimal RI strategy will evolve from not offering CRI to offering CRI. This is because although CRI can improve consumers' willingness-to-pay, the e-tailer will handle more returns, which leads to a high loss. As the salvage value proportion increases, this negative effect will decrease. Therefore, the e-tailer is more willing to offer CRI.

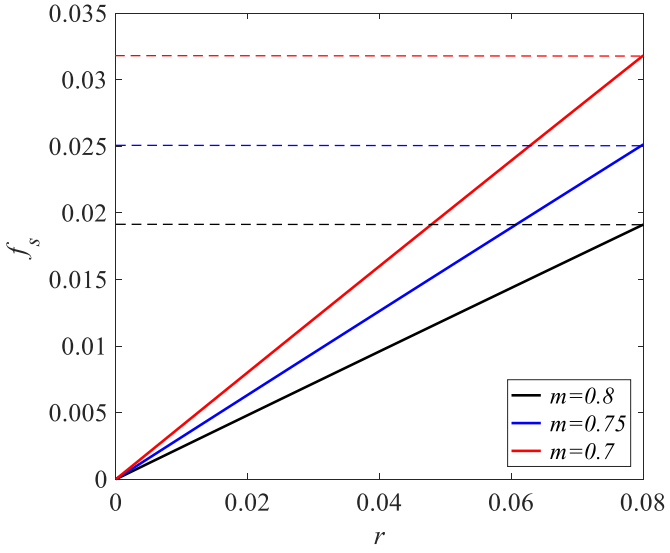


Fig. 5. The impact of m.

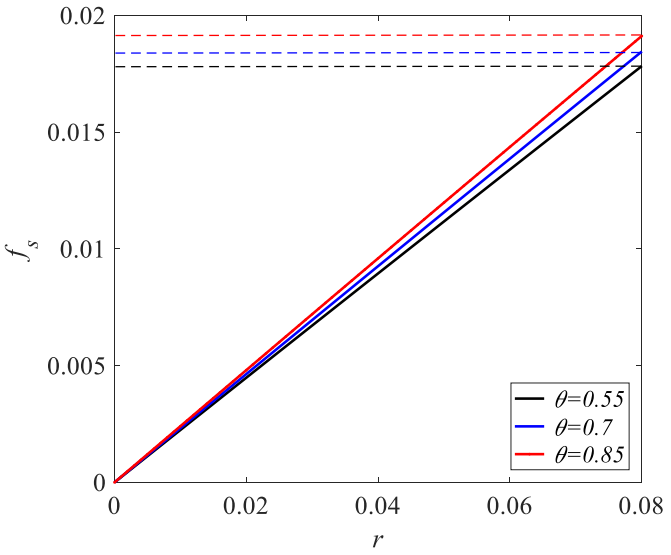


Fig. 6. The impact of θ.

6. Conclusion

It has recently been noted that, in addition to the common return strategy, the question of whether the e-tailer provides CRI also affects consumers' purchase decisions. To reduce the consumer's shopping risk, it can be found that some e-tailer offer CRI. If the e-tailer does not provide this service, then consumers can also purchase RI on their own initiative. To clearly find out the value of RI, we study a two-stage supply chain consisting of a manufacturer and an e-tailer. We then develop a game model to solve the problem of when the e-tailer should offer CRI. Key findings are now summarized:

First, whether the introduction of RI benefits the manufacturer and the e-tailer depends on the latter's RI strategy. When the e-tailer does not offer CRI (RI is available for consumers to purchase), supply chain members will benefit from the existence of RI. This is mainly achieved by earning more profits from the increased demand. Some consumers who would not have bought products will reduce the shopping risk by purchasing RI (the return-freight compensation eases the negative impact of the return-freight fee), and then join the market. When CRI is offered, the impact of RI on supply chain members is not monotonous, which depends on the RI premium incurred by the e-tailer. Specifically, when this premium is high, offering CRI will damage the profits of supply chain members. On the contrary, when this premium is low, the participants can take the advantage of CRI to create more profits.

Second, RI can improve the price level in the market, and this effect is more significant in the scenario of offering CRI. This is reflected in that the wholesale and retail prices in the market with RI are higher than those in the market without RI, respectively; the wholesale and retail prices are the highest in the scenario of offering CRI. In addition, allowing consumers to purchase RI can reduce the consumer's sensitivity to the probability of product fit, while only when the RI premium incurred by the e-tailer is high can CRI achieves the same effect.

Third, the RI premium incurred by the e-tailer and the return-freight compensation play a critical role in the selection of RI strategy for the e-tailer. When this RI premium is high, she should give consumers the opportunity to purchase RI, which is also in line with the traditional common sense. However, the low RI premium does not necessarily encourage the e-tailer to offer CRI. Specifically, when this RI premium is low and the return-fregiht compensation is high, the strategy of offering CRI will dominate. As the return-freight compensation decreases, the e-tailer's optimal RI strategy will evolve from offering CRI to not offering it. In addition, from the perspective of maximizing his own interests, we also find that the manufacturer should also help the e-tailer pay for RI.

From a managerial perspective, our findings provide several useful management insights. For the pricing strategy, the manufacturer and the e-tailer need to make it clear that they should increase their prices after joining the market with RI. In addition, if CRI is given, the wholesale and retail prices should be further increased. For the e-tailer, she needs to grasp some key information to reasonably select the RI strategy, such as the RI premium incurred by her (this premium is in a positive relationship with the e-tailer's historical average return rate of products sold) and the return-freight compensation offered by the insurance company (this compensation is in a positive relationship with the distance between a consumer's address and the return place). For the e-tailer with a high historical average return rate, due to the high risk of product returns and RI premium, it is not recommended that she offers CRI to consumers. For the e-tailer with a low historical average return rate, she may benefit from CRI. Specifically, such an e-tailer can check her own historical sales records to analyze the source of consumers. If most of these consumers come from areas far away from her, she is suitable for offering CRI. Otherwise, this e-tailer should give consumers the opportunity to purchase RI. More importantly, according to the actual observation, the RI premium charged by the insurance company to the e-tailer generally depends on the average return rate in the first three months of the month when the insurance agreement is signed. The dynamic return rate also implies that the e-tailer needs to flexibly adjust her RI strategy rather than blindly adopt one strategy. The e-tailer can also cooperate with the insurance company. To promote new products to occupy the market quickly, for example, the e-tailer can sacrifice some profits to enable the insurance company to provide consumers with the high return-freight compensation or the low RI premium, so as to attract consumers to buy products. In addition, our results also suggest that in the market with RI, consumers' attention to the probability of product fit will be reduced in most cases. This means that the e-tailer who sells inferior products have an opportunity to make profits, because she does not need to provide very detailed product information and picture display. As a result, the e-commerce platform may need to set more stringent review standards. For the manufacturer, he should distinguish the e-tailer's type and pay attention to the actual return-freight compensation. When certain conditions are met, the manufacturer should encourage the e-tailer to offer CRI. Meanwhile, the manufacturer should also fully bear the RI premium for the latter.

Future studies could consider two further aspects. First, we focus on the single channel, and it would therefore be worthwhile and instructive to consider the RI strategy in the dual channel. Second, this paper assumes that consumer demand is certain, and it would therefore be instructive for future work to explore how RI impacts on e-tailers when the demand is uncertain.

Acknowledgments

This work was partly supported by the National Science Foundation of China (Project No. 71871049) and the 111 Project (B16009).

Appendix

Proof of Proposition 1. $\frac{\partial D_N^*}{\partial m} = \frac{s+(1-\theta)c}{4m^2} > 0$, $\frac{\partial D_C^*}{\partial m} = \frac{s+(1-\theta)c}{4m^2} - \frac{(1-m^2)r^2}{8m^2} > 0$. It is clear that $\frac{\partial D_N^*}{\partial m} > \frac{\partial D_C^*}{\partial m}$. $\frac{\partial D_S^*}{\partial m} = \frac{s+(1-\theta)c}{4m^2} - \frac{r-(1-\theta)f_s}{4m^2}$. When $r-(1-\theta)f_s > 0$ (i.e., $r > (1-\theta)f_s$), $\frac{\partial D_N^*}{\partial m} > \frac{\partial D_S^*}{\partial m}$, otherwise, $\frac{\partial D_N^*}{\partial m} \le \frac{\partial D_S^*}{\partial m}$. Therefore, Proposition 1 is proved.

Proof of Proposition 2. $w_S^* - w_N^* = \frac{(1-m)r+\lambda f_s}{2\lambda} > 0$, $w_C^* - w_N^* = \frac{(1-m)^2 r^2}{4\lambda} > 0$; $p_S^* - p_N^* = \frac{3(1-m)r+\lambda f_s}{4m} > 0$, $p_C^* - p_N^* = \frac{3(1-m)^2 r^2}{8m} > 0$. Therefore, Proposition 2 is proved.

Proof of Proposition 3. By Lemmas 1 and 3, $\Pi_S^{M^*} \ge \Pi_N^{M^*}$ and $\Pi_S^{R^*} \ge \Pi_N^{R^*} \Leftrightarrow D_S^* \ge D_N^*$. $D_S^* - D_N^* = \frac{1}{4m}[(1-m)r - \lambda f_s]$. Let $D_S^* - D_N^* \ge 0$, we obtain $f_s \le (1-m)r/\lambda$. By Lemmas 1 and 2, it is clear to find that $\Pi_C^{M^*} \ge \Pi_N^{M^*}$, $\Pi_C^{R^*} \ge \Pi_N^{R^*}$. Therefore, Proposition 3 is proved.

Proof of Proposition 4. $w_S^* - w_C^* = \frac{(1-m)r[1-(1-m)r/2]+\lambda f_s}{2\lambda} > 0$. $p_S^* - p_C^* = \frac{3(1-m)r[1-(1-m)r/2]+\lambda f_s}{4m} > 0$. Therefore, Proposition 4 is proved.

Proof of Proposition 5. By Lemmas 1 and 2, $\Pi_S^{M^*} \ge \Pi_C^{M^*}$ and $\Pi_S^{R^*} \ge \Pi_C^{R^*} \Leftrightarrow D_S^* \ge D_C^*$. $D_S^* - D_C^* = \frac{1}{4m}\{(1-m)r[1-(1-m)r/2] - \lambda f_s\}$. Let $A(r) = (1-m)r[1-(1-m)r/2] - \lambda f_s$, $\frac{\partial A(r)}{\partial r} = (1-m)[1-(1-m)r] \rangle 0$. Therefore, $A(r)$ increases with r, where $r \in [0, s]$. $A(r=0) = -\lambda f_s < 0$, $A(r=s) = (1-m)s[2-(1-m)s]/2 - \lambda f_s$. Let $A(r=s) \le 0$, we get $f_s \ge (1-m)s[2-(1-m)s]/2\lambda$. As a result, when $f_s \ge (1-m)s[2-(1-m)s]/2\lambda$, $D_S^* \le D_C^*$, i.e., $\Pi_S^{M^*} \le \Pi_C^{M^*}$, $\Pi_S^{R^*} \le \Pi_C^{R^*}$. When $f_s < (1-m)s[2-(1-m)s]/2\lambda$ (i.e., $A(r=s) > 0$), let $A(r) = 0$, we obtain $r = \hat{r} = \frac{1-\sqrt{1-2\lambda f_s}}{1-m}$. Thus, when $f_s < (1-m)s[2-(1-m)s]/2\lambda$, if $0 \le r \le \hat{r}$, $D_S^* \le D_C^*$, i.e., $\Pi_S^{M^*} \le \Pi_C^{M^*}$, $\Pi_S^{R^*} \le \Pi_C^{R^*}$; if $\hat{r} < r \le s$, $D_S^* > D_C^*$, i.e., $\Pi_S^{M^*} > \Pi_C^{M^*}$, $\Pi_S^{R^*} > \Pi_C^{R^*}$. Therefore, Proposition 5 is proved.

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